

Mlaby	7048	AgSp1	serine	rm1a	1	PGTTPGTVTGP	DGKPS	E	FIVP	NGAFT	TPGS	IPGS	DGKPIH	VEPAGPGT	TPGAET	GP	DGN	V	L	K		I	L	P	T	P	V	R	P	E	P	Q	P	31		
Mlaby	7048	AgSp1	serine	rm1b	1	PGTTPGTVTGP	DGKPS	E	FIVP	NGAFT	TPGS	IPGS	DGKPIH	VEPAGPGT	TPGAET	GP	DGN	V	L	K		I	L	P	T	P	V	R	P	E	P	Q	P	72		
Mlaby	7048	AgSp1	serine	rm1a	3	PGTTPGVVVTGP	DGQPV	Q	FIVP	QGAFT	TPGS	IPGP	NSKPVH	VRPAGPGT	TPGAET	SDGS	I	Q	S		I	V	L	P	S	T	P	K	P	S	G	..	P	74		
Mlaby	7048	AgSp1	serine	rm1a	12				FIVP	QGAFT	TPGS	IPGP	KGKPVH	VRPAGPGT	TPGAET	SDGS	N	I	L	S		I	V	L	P	S	T	P	K	P	S	R	..	P	57	
Mlaby	7048	AgSp1	serine	rm1a	14				FIVP	QGAFT	TPGS	IPGP	DGKQVH	VRPAGPGT	TPGAET																			36		
Mlaby	7048	AgSp1	serine	rm1a	25																												0			
Mlaby	7048	AgSp1	serine	rm1a	26	PGTTPGVVVTGP	GPG	G	K	P	L																						17			
Mlaby	7048	AgSp1	serine	rm1a	27																												12			
Mlaby	7048	AgSp1	serine	rm1a	28	PGTTPGTVTGP	DGN	P	V		K	FIVP	QGAFT	TPGS	I	S	G	P															53			
Mlaby	7048	AgSp1	serine	rm1a	37																												0			
Mlaby	7048	AgSp1	serine	rm1a	38	PGTTPGVVVTGP	GPG	G	Q	P	L		R	FIVP	QGAFT	TPGS	I	P	G	P														74		
Mg	TRINITY	DN1546	c0	g1	serine	rm	1																										26			
Mg	TRINITY	DN1546	c0	g1	serine	rm	2	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	74
Mg	TRINITY	DN1546	c0	g1	serine	rm	3	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	74
Mg	TRINITY	DN1546	c0	g1	serine	rm	4	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	74
Msag	19329	AgSp1.1	serine	rm	1																												50			
Msag	19329	AgSp1.1	serine	rm	2	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	72		
Msag	19329	AgSp1.1	serine	rm	3	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	73		
Msag	19329	AgSp1.1	serine	rm	4	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	74		
Msag	19329	AgSp1.1	serine	rm	5	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	74		
Msag	19329	AgSp1.1	serine	rm	6	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	Q	G	A	F	T	P	G	S	I	A	G	P	H	G	K	R	V	H	V	K	P	70		
Msag	19329	AgSp1.1	serine	rm2	1																												29			
Msag	19329	AgSp1.1	serine	rm2	16																												0			
Msag	19329	AgSp1.1	serine	rm2	17																												29			
Varen	6869	AgSp1.1	serine	rm2	1	PGTTPGTVTGR	DGKPK	K	FIVP	NGAFT	TPGS	F	P	G	P	D	G	E	P	I	S													75		
Varen	6869	AgSp1.2	serine	rm2	1	PGTTPGVVVTGP	DGQPV	P	V	K	V	I	V	P	R	E	A	F	S	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	74
Varen	6869	AgSp1.1	serine	rm2	2																													21		
Varen	6869	AgSp1.2	serine	rm2	2	PGTTPGVVVTGP	DGQPV	K	V	I	V	P	R	E	A	F	S	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	75	
Varen	6869	AgSp1.2	serine	rm2	3																													14		
Varen	6869	AgSp1.1	serine	rm3	1	PGTTPGIVTGP	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	1	PGTTPGIVTGP	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	2	PGTTPGVVVTNS	DGQPV	E	G	Q	P	I																						74		
Varen	6869	AgSp1.1	serine	rm3	2	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.1	serine	rm3	3	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	3	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	4	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	5	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	6	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	7	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.2	serine	rm3	8	PGTTPGVVVTNS	DGQPV	Q	FIVP	R	G	A	F	R	T	P	G	S	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.1	serine	rm3	9																												37			
Varen	6869	AgSp1.2	serine	rm3	9	PGTTPGIVTGS	DGQPV	K	FIVP	Q	G	A	F	K	T	P	G	F	I	M	G	P	H	G	K	R	V	H	V	K	P	74				
Varen	6869	AgSp1.1	serine	rm3	10	PGTTPGVVVTGP	DGQPV	K	FIVP	Q	G	A	F	R	T	P	G	V	I	P	G	P	H	G	K	R	V	H	V	K	P	74				
Aarg	AgSp1	FC511	serine	rm1	14	PGTTPGTVTGS	DGKPN	K	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	75					
Aarg	AgSp1	FC511	serine	rm2	3b	PGTTPGVVVTGP	DGQPV	K	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	74					
Aarg	AgSp1	FC511	serine	rm2	1	PGTTPGTVTGR	N	G	R	P	K																						74			
Aarg	AgSp1	FC511	serine	rm2	2	PGTTPGVVVTGP	DGQPV	K	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	74					
Aarg	AgSp1	FC511	serine	rm2	3	PGTTPGVVVTGP	DGQPV	K	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	74					
Aarg	AgSp1	FC511	serine	rm2	4	PGTTPGVVVTGP	DGQPV	K	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	74					
Aarg	AgSp1	FC511	serine	rm2	5	VGTTTGP	IVTGP	DGQPV	K	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	75				
Ncruc	58532and13532	AgSp1	serine	rm	87																												61			
Ncruc	58532and13532	AgSp1	serine	rm	88	PGTTPGVVVTGP	DGQPV	Q	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	70					
Ncruc	58532and13532	AgSp1	serine	rm	89	PGTTPGVVVTGP	DGQPV	Q	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	70					
Ncruc	58532and13532	AgSp1	serine	rm	90	PGTTPGVVVTGP	DGQPV	Q	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	74					
Ncruc	58532and13532	AgSp1	serine	rm	91	PGTTPGVVVTGP	DGQPV	Q	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	73					
Ncruc	58532and13532	AgSp1	serine	rm	92	PGTTPGVVVTGP	DGQPV	Q	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	73					
Ncruc	58532and13532	AgSp1	serine	rm	93	PGTTPGVVVTGP	DGQPV	Q	FIVP	Q	G	A	F	T	P	G	S	F	P	G	P	H	G	K	R	V	H	V	K	P	74					
Lcorn	58650	AgSp1.2	serine	rm1	3																												5			
Lcorn	58650	AgSp1.2	serine	rm1	4	PGTTPGVVVTGP																														

Mlaby 7048 AgSp1 serine rm1a 1
Mlaby 7048 AgSp1 serine rm1b 1
Mlaby 7048 AgSp1 serine rm1a 3
Mlaby 7048 AgSp1 serine rm1a 12
Mlaby 7048 AgSp1 serine rm1a 14
Mlaby 7048 AgSp1 serine rm1a 25
Mlaby 7048 AgSp1 serine rm1a 26
Mlaby 7048 AgSp1 serine rm1a 27
Mlaby 7048 AgSp1 serine rm1a 28
Mlaby 7048 AgSp1 serine rm1a 37
Mlaby 7048 AgSp1 serine rm1a 38
Mg TRINITY DN1546 c0 g1 serine rm 1
Mg TRINITY DN1546 c0 g1 serine rm 2
Mg TRINITY DN1546 c0 g1 serine rm 3
Mg TRINITY DN1546 c0 g1 serine rm 4
Msag 19329 AgSp1.1 serine rm 1
Msag 19329 AgSp1.1 serine rm 2
Msag 19329 AgSp1.1 serine rm 3
Msag 19329 AgSp1.1 serine rm 4
Msag 19329 AgSp1.1 serine rm 5
Msag 19329 AgSp1.1 serine rm 6
Msag 19329 AgSp1.1 serine rm2 1
Msag 19329 AgSp1.1 serine rm2 16
Msag 19329 AgSp1.1 serine rm2 17
Varen 6869 AgSp1.1 serine rm2 1
Varen 6869 AgSp1.2 serine rm2 1
Varen 6869 AgSp1.1 serine rm2 2
Varen 6869 AgSp1.2 serine rm2 2
Varen 6869 AgSp1.2 serine rm2 3
Varen 6869 AgSp1.1 serine rm3 1
Varen 6869 AgSp1.2 serine rm3 1
Varen 6869 AgSp1.2 serine rm3 2
Varen 6869 AgSp1.1 serine rm3 2
Varen 6869 AgSp1.1 serine rm3 3
Varen 6869 AgSp1.2 serine rm3 3
Varen 6869 AgSp1.2 serine rm3 4
Varen 6869 AgSp1.2 serine rm3 5
Varen 6869 AgSp1.2 serine rm3 6
Varen 6869 AgSp1.2 serine rm3 7
Varen 6869 AgSp1.2 serine rm3 8
Varen 6869 AgSp1.1 serine rm3 9
Varen 6869 AgSp1.2 serine rm3 9
Varen 6869 AgSp1.1 serine rm3 10
Aarg AgSp1 FC511 serine rm1 14
Aarg AgSp1 FC511 serine rm2 3b
Aarg AgSp1 FC511 serine rm2 1
Aarg AgSp1 FC511 serine rm2 2
Aarg AgSp1 FC511 serine rm2 3
Aarg AgSp1 FC511 serine rm2 4
Aarg AgSp1 FC511 serine rm2 5
Ncruc 58532and13532 AgSp1 serine rm 87
Ncruc 58532and13532 AgSp1 serine rm 88
Ncruc 58532and13532 AgSp1 serine rm 89
Ncruc 58532and13532 AgSp1 serine rm 90
Ncruc 58532and13532 AgSp1 serine rm 91
Ncruc 58532and13532 AgSp1 serine rm 92
Ncruc 58532and13532 AgSp1 serine rm 93
Lcorn 58650 AgSp1.2 serine rm1 3
Lcorn 58650 AgSp1.2 serine rm1 4
Lcorn 58650 AgSp1.2 serine rm1 5
Lcorn 58650 AgSp1.2 serine rm1 6

Q.....T TTKVKGPKQKPLVVPAG.. 52
.....TGPGGQPLNPLGPGSPLRP.....GSPLGPGLNL...ATPATTTPTVPGPGCKPLQIVPAG.. 125
GFN.....FQITPSPTTKPDGQPIQIVPAG.. 98
GK.....QFKNPLPVTQSDGQPIQIVPAG.. 81
.....NPLPVTQSDGQPIQIVPAG.. 36
..... 19
..... 17
.Q.Q.....TTTPEPIPGPNCKPIQIVPAG.. 35
..... 53
.....NPLPVTQPDGQPIQIVPAG.. 19
VP.....ITTT...PIPGPGPEPIQIVPAG.. 96
GF.Q.....SPTTPSHVTKPDGQPIQIVPAG.. 51
GFQS.....PTTPSHVTKPDGQPIQIVPAG.. 99
AFQS.....PTTPSHVTKPDGQPIQIVPAG.. 99
GFQS.....PTTPSHVTKPDGQPIQIVPAG.. 99
G.....F.....QSPTPSQVTKPDGQPIQIVPAG.. 74
PG.....F.....HTPTPSQVTKPDGQPIQIVPAG.. 97
PG.....F.....QSPTPSQVTKPDGQPIQIVPAG.. 98
PG.....F.....RPTPLSQVTKPDGQPIQIVPAG.. 99
PG.....F.....QSPTPSQVTKPDGQPIQIVPAG.. 99
..... 70
GF.Q.....SPTTPRRVTKPACQPIETVAACPG 55
.....YVVPQGAFTTPGSFAGVHGKRVHVPAG.. 28
GF.R.....APTPSRATEPTGQPIQIVTTG.. 53
PM.....GPGSPFGPGNMF.....GP..MR.PGRTTTPSSVEGPDGEP I Q I E P A G .. 117
GF.....DFHTPSHVTKGDKGKPIQIVPAG.. 98
NG.A.....FTTPGSFPGPDGEPISVEPACPG 47
GF.....DFHTPSHVTKGDKGKPIQIVPAG.. 99
.....PVKGDGQPIQIVPACPG 31
RY.....RFQTPSRVTKPDGEP I E T V P A G .. 98
GV.....HFETTKAITEPDGQPIVIVPAG.. 98
GF.....QFQTPKQITEPDGQPIRIIPAG.. 98
GF.....QFQTPSRVTKPDGQPIETVPAG.. 98
.....PG 2
GY.....QFQTPQKITEPDGQPIETIPAG.. 98
GF.....QFQTPKHITEPDGQPIRIIPAG.. 98
GY.....QFQTPKKITEPDGQPIETIPAG.. 98
GF.....QFQTPKHITEPDGQPIRIIPAG.. 98
GY.....QFQTPKKITEPDGQPIETIPAG.. 98
GF.....QSQTPTNTGTGTCQPIQIIPAG.. 98
GF.R.....FPTTPRRVPMNPDGQPIQIIPAG.. 61
GF.....QSQTPTNTKTGTCQPIQIIPAG.. 98
GFQ..... 77
GG.....KKGTGENPANSNNAIGP..LQ.TTSTTTPTPTSISGPDGQPMQVVPAG.. 120
G.....F.....QTPKSVTKPDGQPIQVVPAG.. 98
GF.....PTF.....QFQTPRPPIKGPCKKRIQVVPAG.. 101
GL.....QFQTPKPIKGPAGKPLQIIPAG.. 98
GF.....QFQTPRPITAPGCKPIQVVPAG.. 98
G.....F.....QF.....STPGPVPPQPDGQPIQIVPAG.. 98
.....PANPTTPTDVKGPQCGPVMIIPAG.. 99
.....HTPSPTKPDGKPIQIVPAG.. 81
.....GSGPGF..QF.....QTPQPVTKPDGQPIQIVPAG.. 98
.....GSGPGF..QF.....QTPQPVTKPDGQPIQIVPAG.. 98
G.....FPTTPKG.....SGPGF..QF.....QTPQPVTKPDGQPIQIVPAG.. 109
.....PGF..NF.....ETPRPAPQ.SGGPIQIVPAG.. 97
.....PGY..NF.....PTPRPAPQ.SGGPIQIVPAG.. 97
.....QP.A.....NPTTPTDVRGPGQSGPIPIVPAG.. 99
.F.Q.....FQTPQPITKPGCGQPIQIVPAG.. 28
.....G.....SRPGF..QF.....QTPQPITKSGGQPIQIVPAG.. 98
.....PG 2
OG.A.....FTTSGSTPGPNCKPIHVKPAGPG 30

Mlaby 7048 AgSp1 serine rm1a 1		52
Mlaby 7048 AgSp1 serine rm1b 1		125
Mlaby 7048 AgSp1 serine rm1a 3		98
Mlaby 7048 AgSp1 serine rm1a 12		81
Mlaby 7048 AgSp1 serine rm1a 14		36
Mlaby 7048 AgSp1 serine rm1a 25		19
Mlaby 7048 AgSp1 serine rm1a 26		17
Mlaby 7048 AgSp1 serine rm1a 27		35
Mlaby 7048 AgSp1 serine rm1a 28		53
Mlaby 7048 AgSp1 serine rm1a 37		19
Mlaby 7048 AgSp1 serine rm1a 38		96
Mg TRINITY DN1546 c0 g1 serine rm 1		51
Mg TRINITY DN1546 c0 g1 serine rm 2		99
Mg TRINITY DN1546 c0 g1 serine rm 3		99
Mg TRINITY DN1546 c0 g1 serine rm 4		99
Msag 19329 AgSp1.1 serine rm 1		74
Msag 19329 AgSp1.1 serine rm 2		97
Msag 19329 AgSp1.1 serine rm 3		98
Msag 19329 AgSp1.1 serine rm 4		99
Msag 19329 AgSp1.1 serine rm 5		99
Msag 19329 AgSp1.1 serine rm 6		70
Msag 19329 AgSp1.1 serine rm2 1	TTPGVVTGPDGQPVRYPVPPQGAFTTPGSFAGLHGK.....	90
Msag 19329 AgSp1.1 serine rm2 16		28
Msag 19329 AgSp1.1 serine rm2 17		53
Varen 6869 AgSp1.1 serine rm2 1		117
Varen 6869 AgSp1.2 serine rm2 1		98
Varen 6869 AgSp1.1 serine rm2 2	TTPGIETDSDGNIKKVVVPTTT.....	69
Varen 6869 AgSp1.2 serine rm2 2		99
Varen 6869 AgSp1.2 serine rm2 3	TTLGVVTGPDGQPVKPRAVTEPDGEP.....IQIIPAG	64
Varen 6869 AgSp1.1 serine rm3 1		98
Varen 6869 AgSp1.2 serine rm3 1		98
Varen 6869 AgSp1.2 serine rm3 2		98
Varen 6869 AgSp1.1 serine rm3 2		98
Varen 6869 AgSp1.1 serine rm3 3	TTPGVVTGPDGQPVFIVPRGAFRT.....	27
Varen 6869 AgSp1.2 serine rm3 3		98
Varen 6869 AgSp1.2 serine rm3 4		98
Varen 6869 AgSp1.2 serine rm3 5		98
Varen 6869 AgSp1.2 serine rm3 6		98
Varen 6869 AgSp1.2 serine rm3 7		98
Varen 6869 AgSp1.2 serine rm3 8		98
Varen 6869 AgSp1.1 serine rm3 9		61
Varen 6869 AgSp1.2 serine rm3 9		98
Varen 6869 AgSp1.1 serine rm3 10		77
Aarg AgSp1 FC511 serine rm1 14		120
Aarg AgSp1 FC511 serine rm2 3b		98
Aarg AgSp1 FC511 serine rm2 1		101
Aarg AgSp1 FC511 serine rm2 2		98
Aarg AgSp1 FC511 serine rm2 3		98
Aarg AgSp1 FC511 serine rm2 4		98
Aarg AgSp1 FC511 serine rm2 5		99
Ncruc 58532and13532 AgSp1 serine rm 87		81
Ncruc 58532and13532 AgSp1 serine rm 88		98
Ncruc 58532and13532 AgSp1 serine rm 89		98
Ncruc 58532and13532 AgSp1 serine rm 90		109
Ncruc 58532and13532 AgSp1 serine rm 91		97
Ncruc 58532and13532 AgSp1 serine rm 92		97
Ncruc 58532and13532 AgSp1 serine rm 93		99
Lcorn 58650 AgSp1.2 serine rm1 3		28
Lcorn 58650 AgSp1.2 serine rm1 4		98
Lcorn 58650 AgSp1.2 serine rm1 5	TTPGVVTGADGQPVK.....	17
Lcorn 58650 AgSp1 2 serine rm1 6	TTPGAK.....	36

 non-conserved
 similar
 $\geq 50\%$ conserved